Patrick Banner

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Education

Ph.D. in Physics, University of Maryland at College Park, August 2025
Advisors: Trey Porto and Steve Rolston
Doctoral Dissertation: "Rydberg Atoms, Photonic Devices, and Graduate Student Mental Health: Embracing Complexity and Care in Physics"

B.S. in Physics, *Summa Cum Laude* (class Valedictorian), Denison University, May 2018
Undergraduate Research Advisor: Steve Olmschenk
Undergraduate Thesis: "Electro-optic Modulation of Infrared Light for Laser Cooling of La III"

Experience

Visiting Assistant Professor, Swarthmore College (August 2025 – present)

Courses taught: General Physics I (Phys 3; introductory mechanics), General Physics II (Phys 4; introductory E&M), Quantum Mechanics (Phys 107)
Research: noise and infidelity in fiber-based quantum networks
Mentoring: one senior engineering capstone project (three students), two research students

Graduate Research Assistant, University of Maryland (August 2018 – August 2025)

Teaching: co-developed and co-instructed "The Quantum Wave" (Phys 137) about impacts of quantum technology on society; TA for Modern Physics (Phys 371); grader for graduate AMO seminar (Phys 721); individual tutor for quantum mechanics (Phys 401) and graduate NMR methods (Biochem 669E)
Research in multiple areas: (1) applications of Rydberg ensembles for quantum networking applications; (2) new algorithms for single-photon avalanche diodes; (3) noise and infidelity in fiber-based quantum networks; (4) graduate student educational experiences and graduate student mental health
Mentored three undergraduates in research
Won grant from American Physical Society Forum on Education Mini-Grant Program
Obtained certification from University Teaching and Learning Program, Associate Level
Led the Physics Graduate Student Mental Health Task Force
Writing Fellow in the Graduate School Writing Center

Research and Teaching Assistantships, Denison University (August 2015 – May 2018)

Teaching: Lab assisting for intro physics for non-majors (Phys 121, 122) and intro astronomy (Astro 100); individual tutoring for intro physics for non-majors (Phys 121), intro physics for majors (Phys 125, 126, 127)
Research: laser trapping and cooling of La III for quantum networking
Volunteer for physics department outreach to local elementary school
Editor-in-chief, *Adytum* (Denison University yearbook)

Publications

Refereed Journal Articles:

P. R. Banner, S. L. Rolston, and J. W. Britton, "BIFROST: A First-Principles Model of Polarization Mode Dispersion in Optical Fiber," *Physical Review Applied* (accepted) (2026), DOI: 10.1103/xgqr-rlmf.

P. R. Banner, K. O'Brien, and C. Turpen, "Co-Designing for Graduate Student Mental Health: An Example of Participatory Research Methodologies in PER," *PERC Proceedings 2025*, DOI: 10.1119/perc.2025.pr.Banner.

D. Kurdak, **P. R. Banner,** Y. Li, S. R. Muleady, A. V. Gorshkov, S. L. Rolston, and J. V. Porto, "Enhancement of Rydberg Blockade via Microwave Dressing," *Physical Review Letters* 134, 123404 (2025), DOI: 10.1103/PhysRevLett.134.123404.

P. R. Banner, D. Kurdak, Y. Li, A. Migdall, J. V. Porto, and S. L. Rolston, "Number-State Reconstruction with a Single Single-Photon Avalanche Detector," *Optica Quantum* 2, **2** (2024), DOI: 10.1364/OPTICAQ.504308.

P. R. Banner, D. Kurdak, Y. Li, A. Migdall, J. V. Porto, and S. L. Rolston, "Number-State Reconstruction with a Single SPAD," Quantum Sensing, Imaging, and Precision Metrology II (Conference Proceedings) **12912** (2024), DOI: 10.1117/12.3008269.

S. Olmschenk, **P. Banner,** J. Hankes, and A. Nelson, "Optogalvanic spectroscopy of the hyperfine structure of the $5p^6 5d^2 D_{3/2,5/2}$ and $5p^6 4f^2 F^o_{5/2,7/2}$ levels of La III," *Phys. Rev. A* **96**, 032502 (2017).

Journal Articles in Prep:

P. R. Banner, K. O'Brien, C. Turpen, "What Research Reveals about How to Support Graduate Student Mental Health" (in review for *Physics Today*)

D. Kurdak, Y. Li, **P. R. Banner,** J. V. Porto, and S. L. Rolston, "High-Fidelity Microwave-Polarization Control in a Rydberg-Ensemble Experiment" (in peer review, submitted to *Phys Rev Applied*)

Popular Media:

P. R. Banner, "Solving the skyscraper," news feature on the Denison University website (April 28, 2016), https://denison.edu/feature/66750.

P. R. Banner, "Cool physics," news feature on the Denison University website (Feb. 19, 2016), https://denison.edu/academics/physics/feature/64820.

Fellowships and Awards (Selected)

ARCS Foundation Metro-Washington Chapter 55th Anniversary Named Scholar 2024-2025
Charles T. Husar Fellowship in Physics 2023-2024
National Science Foundation Graduate Research Fellowship 2018-2023
Provost's Academic Excellence Award, Denison University April 2018
Distinguished Leadership Award, Denison University April 2018

Presentations and Posters

Invited:

P. R. Banner, K. O'Brien, and C. Turpen, "Mental Health of Physics and Astronomy Graduate Students: Results from a National Survey", talk at AAPT (August 5, 2025).

P. R. Banner, D. Kurdak, Y. Li, A. Migdall, J. V. Porto, and S. L. Rolston, "Number-state reconstruction with a single SPAD", talk given at SPIE Photonics West (January 31, 2024).

Contributed:

P. R. Banner, K. O'Brien, and C. Turpen, "Co-Designing for Graduate Student Mental Health: An Example of Participatory Research in PER", poster at Physics Education Research Conference (August 7, 2025).

D. Kurdak, **P. R. Banner,** Y. Li, J. V. Porto, and S. L. Rolston, "Arbitrary Microwave Polarization Control in a Rydberg Ensemble Experiment", poster presented at DAMOP (June 19, 2025).

P. R. Banner, Y. Li, D. Kurdak, S. R. Muleady, A. V. Gorshkov, S. L. Rolston, and J. V. Porto, "Demonstration and Modeling of Microwave-Enhanced Rydberg Interactions", talk at DAMOP (June 18, 2025).

Y. Li, D. Kurdak, **P. R. Banner,** S. L. Rolston, and J. V. Porto, "Tailoring Rydberg Interactions via Microwave Dressing", poster presented at DAMOP (June 18, 2025).

P. R. Banner, S. L. Rolston, and J. Britton, "Modeling of Polarization Dynamics in Optical Fibers for Quantum Networking", poster presented at DAMOP (June 17, 2025).

P. R. Banner, "How to Make the Quantum Internet", talk at ARCS Foundation University of Maryland Site Visit (April 15, 2025).

P. R. Banner, S. L. Rolston, and J. Britton, "Simulating Optical Fibers for Quantum Networks", NIST monthly Quantum Optics Meeting (April 1, 2025)

P. R. Banner, S. L. Rolston, and J. Britton, "BIFROST: A Tool for Simulating Optical Fibers for Quantum Networking", poster presented at SCY-QNET Quantum Networking Town Hall (February 27-28, 2025).

P. R. Banner, A. Ehrenberg, Z. Steffen, E. Bennewitz, K. O'Brien, and C. Turpen, "Activities and Survey Results from a Graduate Student Mental Health Task Force in Physics", poster presented at the Physics Education Research Conference (July 10, 2024).

D. Kurdak, **P. R. Banner**, Y. Li, J. V. Porto, and S. L. Rolston, "Improvement of a Rydberg-based Single-photon Source with Microwave Dressing", talk given at DAMOP (June 7, 2024).

P. R. Banner, D. Kurdak, Y. Li, S. L. Rolston, and J. V. Porto, "Number-state reconstruction with a single SPAD", talk given at DAMOP (June 5, 2024).

P. R. Banner, D. Kurdak, Y. Li, S. L. Rolston, and J. V. Porto, "Control of Rydberg Interactions via Microwave Dressing", poster presented at DAMOP (June 4, 2024).

C. Turpen, **P. R. Banner,** R. Dalka, E. Fiala, and B. Rezende, "Disrupting competitive cultures in science through deliberate instructional practices in a Modern Physics course", talk given at AAPT (July 18, 2023).

C. Turpen and **P. R. Banner,** "Disrupting competitive cultures in science through deliberate instructional practices," talk given at the UMD Innovations in Teaching and Learning Conference (May 11, 2022)

P. R. Banner, D. Kurdak, J. V. Porto, and S. L. Rolston, "Prospects for Generation and Measurement of Arbitrary Photon Fock States," talk given at DAMOP (June 3, 2021).

J. Hankes, Amanda Nelson, **Patrick Banner,** and S. Olmschenk, "Towards trapping and laser cooling Ba and La ions," poster presented at DAMOP (June 8, 2017).

A. Nelson, J. Hankes, **P. R. Banner,** and S. Olmschenk, "Optogalvanic spectroscopy of lanthanum hyperfine structure," poster presented at DAMOP (June 7, 2017).

+ 20 talks in local communities, group meetings, journal clubs, and high school outreach

Research Experience

Hardware skills:

Laser systems, control theory, optical cavities, general optical setups, atomic vapor cells, acousto-optic and electro-optic devices, PCBs, CNC machining, soldering, rf and microwave electronics and sources.

Software and coding skills:

Arduino, Autodesk Fusion 360, Autodesk Inventor, C++ (including OpenMP), EAGLE, H5 data format, KiCad, LabView, LaTeX, Mathematica, MatLab, OSLO, Python (including NumPy, SciPy, Pandas, H5, Statsmodels, Pingouin, and Networkx packages), Raspberry Pi, SimIon.

Undergraduates Mentored:

Deven Bowman ‘24 (primary mentor, May 2021-December 2023) (now: grad student at Stanford)
Evan McClintock ‘24 (secondary mentor, May 2023-May 2024) (now: employed at NAVAIR)
Jade LeSchack ‘25 (secondary mentor, Jan 2022-May 2023) (now: grad student at USC)

Research Assistant, Rubidium Rydberg Lab (September 2018 – July 2025)

Advisors: Trey Porto and Steve Rolston
Research Goals: Harness strong interactions between Rydberg atoms for applications in studies of few-body physics with photons, quantum networking, and generation of nonclassical states of light.

- Developed full computational model for single spin-wave decoherence dynamics in Rydberg ensembles using Python (esp. QuTiP and ARC packages) and Mathematica, which we used to improve Rydberg spin-wave lifetimes, especially by finding the “magic” lattice wavelength
- Originated full model for characterizing single-photon avalanche detector
- Algorithmically optimized computation of single-photon avalanche detector properties in Python and C++
- Developed Floquet calculation notebook for understanding details of Rydberg-Rydberg interactions
- Realized rf hardware source for interrogating Rydberg atoms with microwaves with controllable power, polarization, and frequency
- Generated extensive C++ code for processing of H5 time-tagged photodiode data files
- Designed and created programmable three-port RF source for setting and generating acousto-optic modulation frequencies
- Performed Monte Carlo simulations of spin-wave readout to investigate efficiency of single-photon generation
- Developed improved probability models for photon number-state tomography using the expectation-maximization and other maximum likelihood algorithms
- Engineered various printed circuit boards using KiCad, including a board to monitor and regulate temperatures together with a Raspberry Pi
- Used CAD software and CNC milling to create improved magnetic bias field coils
- Provided 20+ tours for local conferences (e.g. CuWiP 2020) and university guests (various visiting scholars, consultants, and government officials)

Research Assistant, Britton Lab (January 2023 – July 2025)

Advisor: Joe Britton
Research Goals: Investigate the properties of optical fibers, especially their polarization properties, and apply experimentally measured properties to quantum networking applications.

- Developed Python codebase for optical fiber modeling based on material properties and previously developed “hinge model”
- Conducted extensive literature review of classical characterizations of optical fibers and classical telecommunication networks

- Assisted with development of high-speed polarimeter, including optical setup, electronics, and software GUI

Principal Investigator, National Survey of Physics Graduate Student Mental Health
(August 2023 – July 2025)
Faculty Colleagues: Chandra Turpen
Research Goals: Quantitatively investigate physics and astronomy graduate student mental health, especially impostor phenomenon, on a national scale.
- Wrote proposal and received grant ($1,800) from American Physical Society Forum on Education Mini-Grant program
- Conducted ~5 validation interviews for survey items on mental health
- Developed and led co-design session with 8 physics graduate students to elicit student feelings about graduate school, mental health, and impostor phenomenon
- Created ~100-item mental health survey combining original questions with clinically and psychometrically validated mental health instruments
- Developed analysis plan for a variety of survey data types, including quantitative semi-continuous, ordinal, categorical, and binary variables as well as qualitative responses
- Analyzed survey data using Google Colab and Python (especially Pandas, NumPy, SciPy, Statsmodels, Pingouin)
- Conducted factor analysis on original survey items to determine statistical structure
- Conducted extensive literature review of graduate student mental health, impostor phenomenon as a psychological construct, and mental health survey instruments
- Reported initial results to APS Fed in 7-page infographic document, publicly available

Undergraduate Research Assistant, Ion Quantum Optics Lab (May 2016 – May 2018)
Research Goals: Trap and laser cool double-ionized lanthanum and explore its telecom electronic transitions in the context of quantum networks.
- Developed hardware and software for multi-frequency rf source to drive multiple electro-optic modulators for use in laser cooling
- Created high-resolution imaging system for ion cloud imaging
- Constructed small lab parts, such as optical mounting equipment, using CNC milling and other machine shop skills

Service and Other Activities

UMD Physics and Astronomy Mental Health Task Force (March 2019-present)
- Originated, constructed, and launched Physics Graduate Student Guide to promote communication and community among graduate students, publicly available
- Participated in American Physical Society panel of graduate student leaders in mental health ("Student to Student: Advocacy and Peer Initiatives to Strengthen Graduate Student Mental Health", Nov 2021, available on YouTube)
- Developed and administered 5 surveys of graduate student mental health to physics and astronomy departments, with IRB approval (2019, 2021 and 2023)
- Analyzed survey data from 6 instruments and other questions using various statistical methods, including principal component analysis and basic network analysis, with Python's Pandas, NumPy, SciPy, and Networkx packages
- Developed 4 reports and 5 presentations (Frontiers and Foundations of Physics 2020, 2021, 2022 and Astronomy BANG! 2019 and 2021) to communicate results of mental health surveys and to advocate for mental health awareness in the department

- Hosted Physics Colloquium speaker Phil Buhlmann (Nov 2022) and organized various meetings and lunches for department discussion about mental health
- Promulgated data and results to departmental stakeholders
- Advocated for departmental policy changes, including overhaul of requirements for first-years, which began Fall 2021
- Supported and coordinated activities with the physics Graduate Committee and Physicists of Underrepresented Genders
- Performed literature searches in the fields of mental health instruments and investigations of mental health in academia

Writing Fellow, UMD Graduate School Center for Written and Oral Communication

(August 2023-present)

- Held 30 weekly one-hour sessions with clients to discuss and develop writing
    - Types of writing included thesis chapters, job and fellowship application materials, journal articles, and coursework
    - Fields included physics, astronomy, biology, architecture, education, and others
- Attended monthly professional development trainings on topics ranging from job applications to the use of AI in writing
- Coordinated writing-in-the-disciplines project about the use of citations in physics; resulting guide publicly available
- Advertised Center and its services to physics and astronomy graduate students and faculty